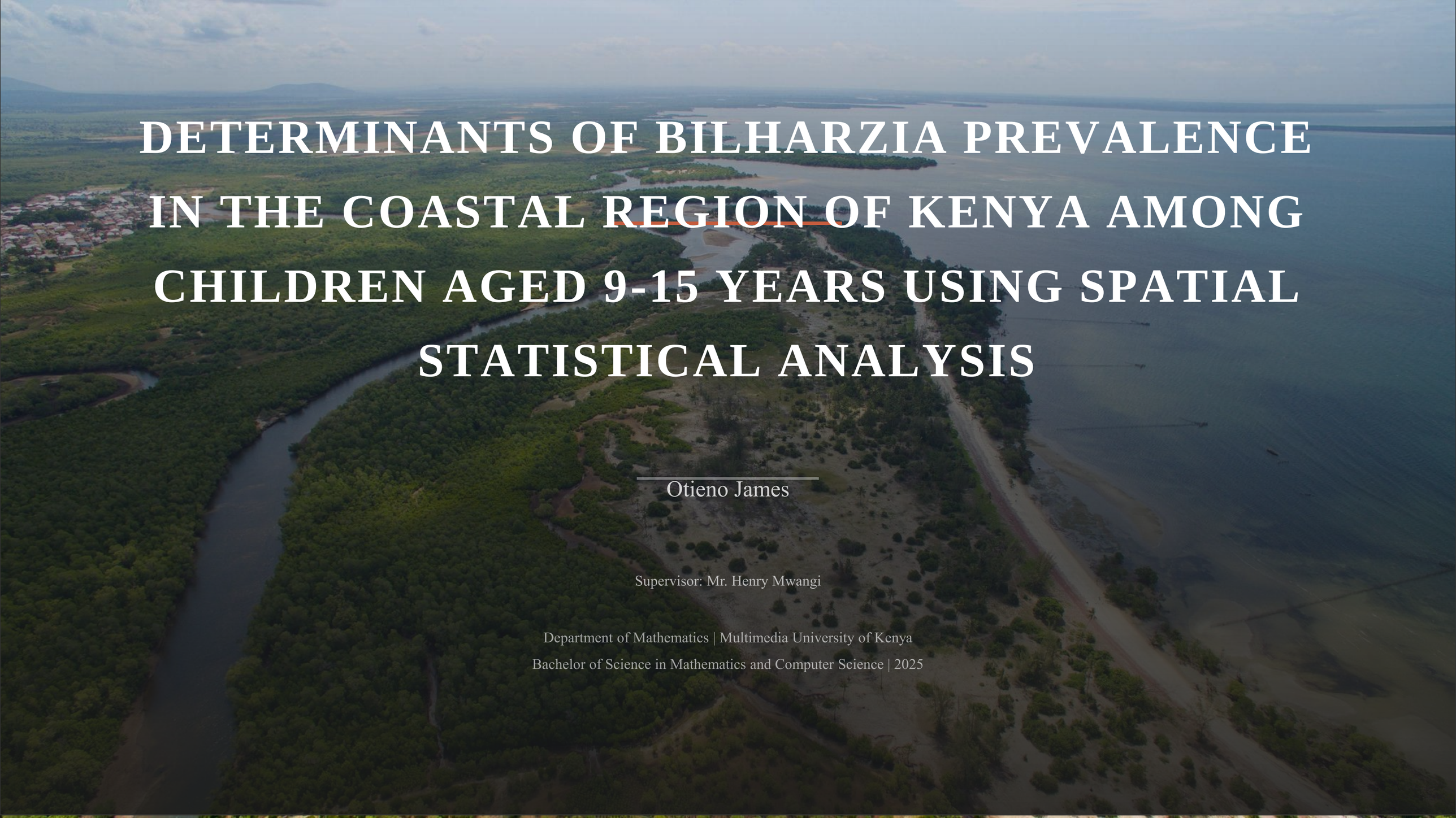


DETERMINANTS OF BILHARZIA PREVALENCE IN THE COASTAL REGION OF KENYA AMONG CHILDREN AGED 9-15 YEARS USING SPATIAL STATISTICAL ANALYSIS

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PRESENTATION OVERVIEW

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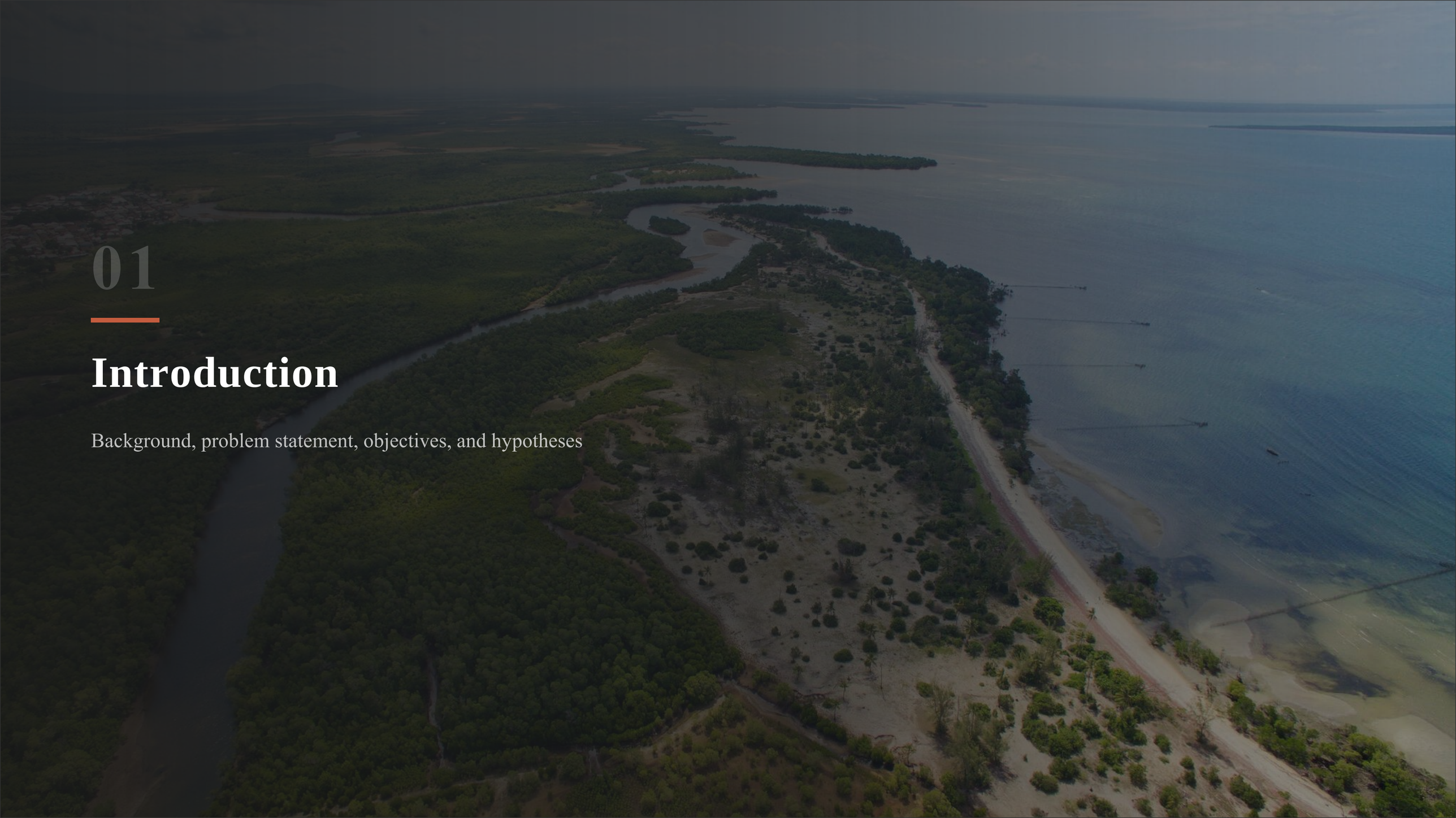
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01

Introduction

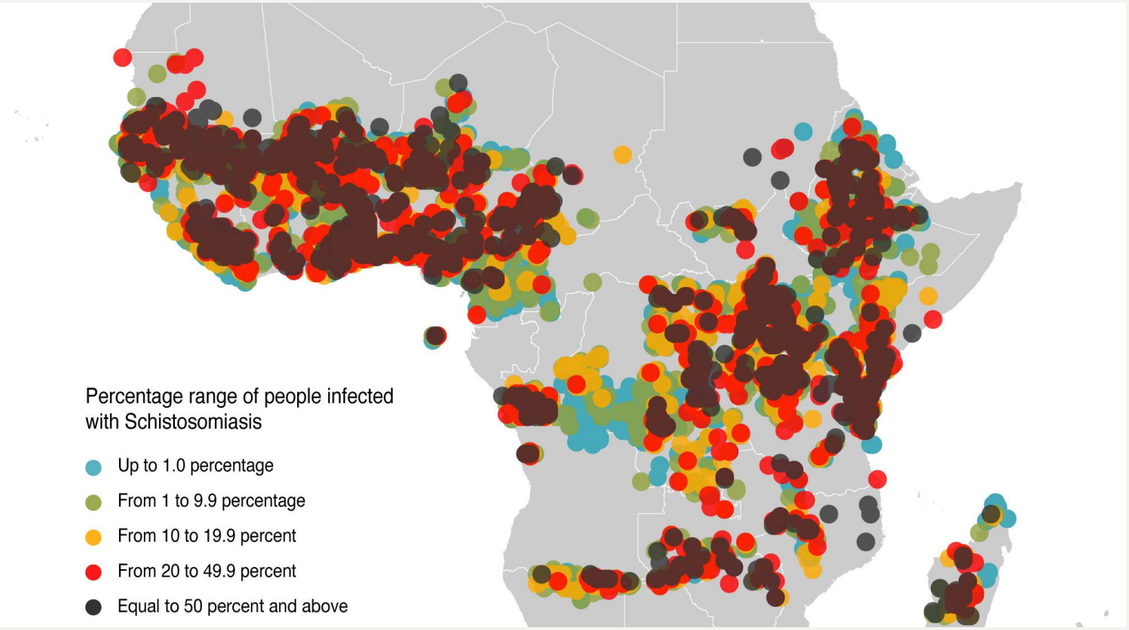
Background, problem statement, objectives, and hypotheses

Background Information

Bilharzia (Schistosomiasis) is a tropical parasitic disease caused by *Schistosoma* worms, remaining a major public health concern in sub-Saharan Africa.

In Kenya, **over 2.5 million people** are at risk of infection, with the coastal region experiencing persistent transmission due to ecological conditions favoring snail habitats.

Children aged **9–15 years** are a key demographic due to frequent water contact through swimming, fishing, and washing activities.



Schistosomiasis prevalence distribution across Africa (Global Atlas of Helminth Infections)

2.5M+
People at risk in Kenya

6.9%
Average prevalence (school-age children)

~20%
Peak prevalence in Tana River County

250M+
People needing preventive chemotherapy globally

Sources: WHO (2023); Kepha et al. (2023); Mutono et al. (2025)

Statement of the Problem

Despite national efforts including **periodic mass drug administration** and health education campaigns, Bilharzia persists among children aged 9–15 in Kenya's coastal region.

Existing studies focus on **overall prevalence** without fully examining **spatial distribution and clustering patterns** across sub-counties or wards.

Spatial heterogeneity suggests some areas experience significantly higher prevalence due to local environmental conditions, water usage, and sanitation access.

Traditional non-spatial approaches fail to capture locational dependence, leading to sub-optimal targeting of control measures.

Research Need

There is a critical need for **spatial statistical analysis** that can:

- Identify **high-prevalence areas** at sub-county level
- Quantify **spatial associations** between environmental and demographic factors
- Inform **targeted public health interventions** and resource allocation

Scope of the Study

- **Study Area:** Coastal region of Kenya, all sub-counties with available data
- **Target Population:** Children aged 9–15 years
- **Spatial Unit:** Sub-county level analysis
- **Data:** Aggregated demographic and environmental variables linked to geographical shapefiles

Research Objectives & Hypotheses

GENERAL OBJECTIVE

To examine the **spatial distribution and spatial determinants** of Bilharzia prevalence among children aged 9–15 years in the coastal region of Kenya.

SPECIFIC OBJECTIVES

1. To analyze **spatial variation** in Bilharzia prevalence across sub-counties
2. To examine **spatial distribution of demographic characteristics** (age, gender) and their association with prevalence
3. To assess **spatial association between environmental factors** (water source, toilet availability) and prevalence
4. To assess the presence and strength of **spatial autocorrelation** in Bilharzia prevalence among sub-counties

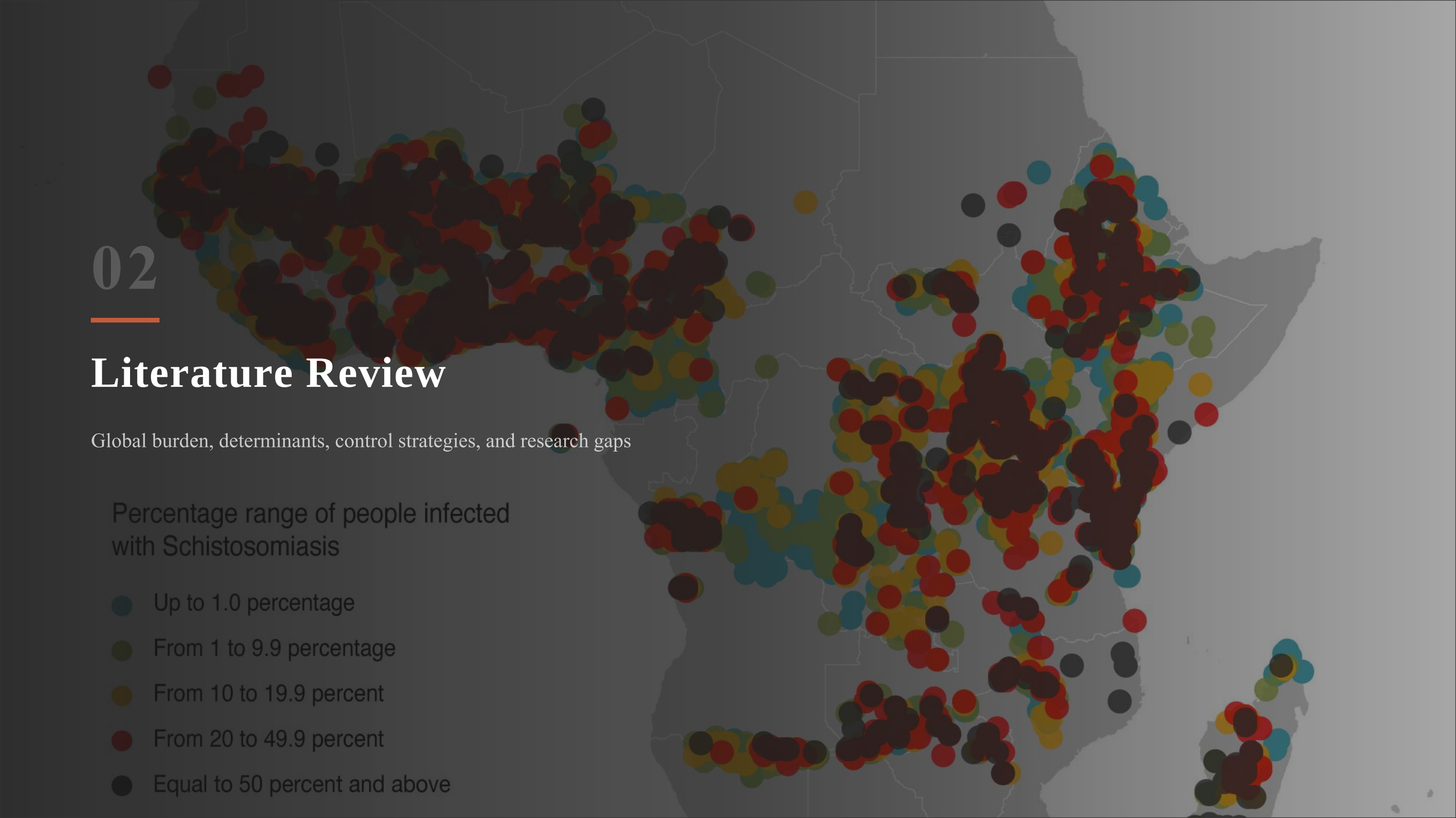
HYPOTHESES

Code	Null Hypothesis (H0)	Alternative Hypothesis (H1)
H01/H11	No significant spatial variation in prevalence	Significant spatial variation exists
H02/H22	No spatial association between demographics and prevalence	Significant spatial association exists
H03/H33	No spatial association between environmental factors and prevalence	Significant spatial association exists
H04/H44	No significant spatial autocorrelation in prevalence	Significant spatial autocorrelation exists

02

Literature Review

Global burden, determinants, control strategies, and research gaps



Schistosomiasis: Global & Regional Burden

Schistosomiasis is caused by trematode worms of the genus *Schistosoma*. Infection occurs through skin contact with freshwater contaminated by cercariae released from infected snails.

The WHO estimates **250+ million people** required preventive chemotherapy in 2021, with over **90% of the burden** in sub-Saharan Africa.

250M+

People requiring preventive chemotherapy (WHO, 2023)

90%

Of global burden in sub-Saharan Africa

70%+

S. mansoni prevalence among school-aged children near Lake Victoria, western Kenya (Odiere et al., 2012)

Impact on Children Aged 9–15

Children aged 9–15 are most vulnerable due to:

- Increased water contact (swimming, fishing, herding)
- Partial immunity insufficient to prevent reinfection
- Chronic effects: **anemia, stunted growth, impaired cognition, reduced school attendance**

Endemic Areas in Kenya

- **Lake Victoria basin** — highest prevalence
- **Coastal region** — persistent transmission
- **Irrigation areas** of central Kenya

Marked differences between counties, sub-counties, and neighboring communities due to local environmental and social conditions.

Sources: WHO (2023); Odiere et al. (2012); Chieng et al. (2006)

Determinants of Bilharzia Prevalence

Environmental Determinants

• **Proximity to freshwater** (rivers, lakes, irrigation canals, ponds) significantly increases infection risk

• **72.5% prevalence** in irrigation areas of Kirinyaga County (Chieng et al., 2006)

• Prevalence **decreases with distance** from Lake Victoria (Woodhall et al., 2013)

• **Poor sanitation** sustains transmission by contaminating water bodies

Demographic Determinants

• **Age:** Infection rates increase through childhood, peaking during adolescence due to cumulative exposure

• **Gender:** Differences reflect culturally defined water-contact roles (fishing, water collection)

• **Socio-economic factors:** Limited safe water and sanitation = higher infection risk (Ng'ang'a et al., 2016)

Control Strategies

Mass Drug Administration (MDA) using praziquantel is the primary control strategy. WHO recommends periodic treatment of at-risk populations, especially school-aged children.

However, **reinfection often occurs shortly after treatment** in Kenya, suggesting uniform strategies may fail to address localized transmission hotspots.

Spatial Epidemiology Approaches

Spatial epidemiology incorporates geographical location into analysis, recognizing that nearby observations may be more similar.

Key techniques: disease mapping, spatial autocorrelation measures, spatial regression models.

Failure to account for spatial autocorrelation may result in **biased estimates** and misleading inference.

Research Gaps

Despite extensive research on schistosomiasis in Kenya, several important gaps remain in the literature:

Chieng et al. (2006)

Reported **72.5% prevalence** among school-aged children in Kirinyaga County irrigation areas but **did not examine spatial variation** or clustering of cases across villages.

Ng'ang'a et al. (2016)

Documented schistosomiasis prevalence and co-infection patterns in Migori County but relied mainly on **descriptive statistics** without spatial statistical techniques.

Odiere et al. (2012)

Reported high prevalence associated with proximity to Lake Victoria in Mbita District but **did not formally test for spatial autocorrelation** or model spatial dependence.



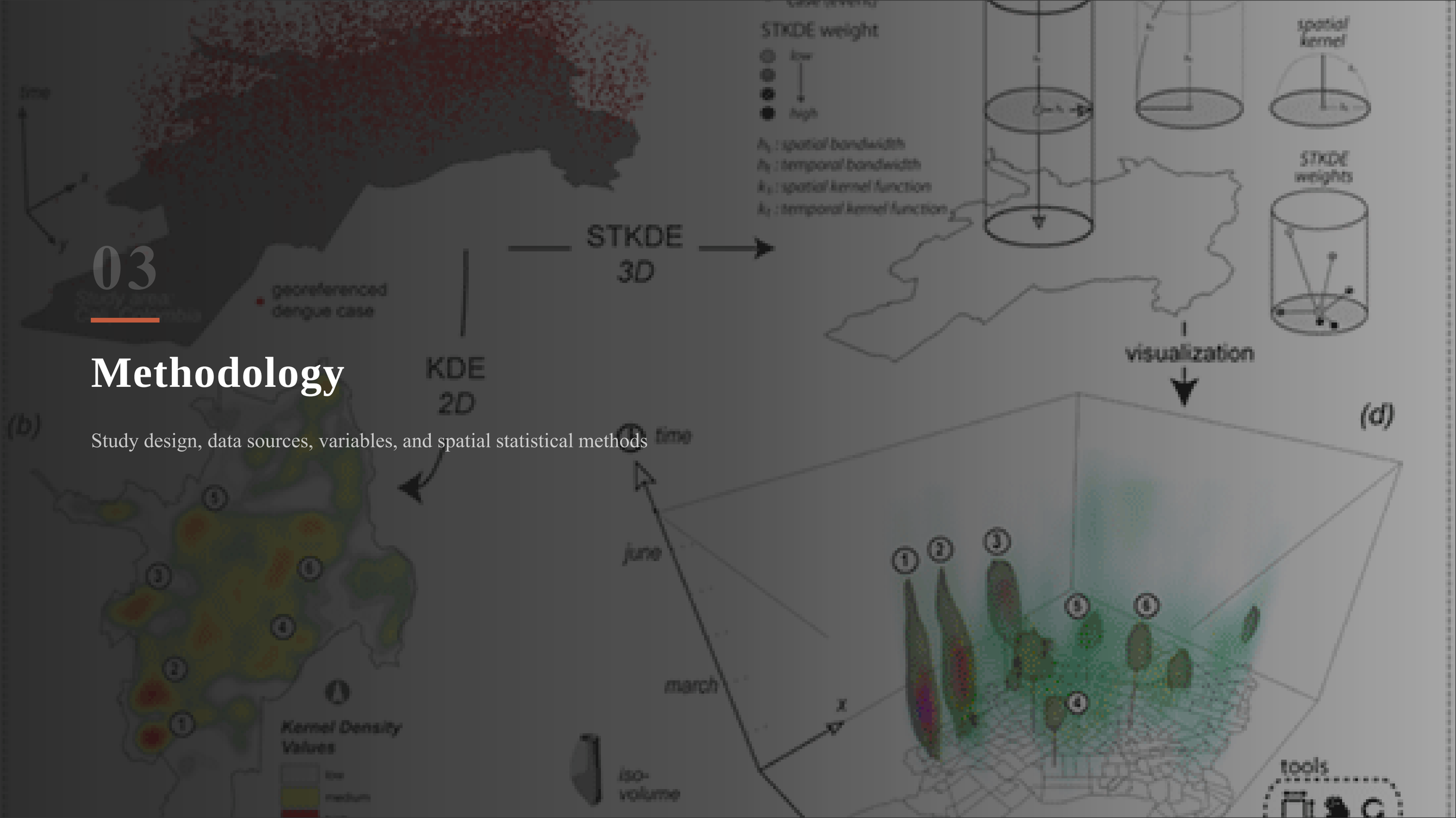
This Study's Contribution

This study addresses these gaps by applying spatial statistical methods to analyze spatial variation, clustering, and determinants of Bilharzia prevalence among children aged 9–15 years in the coastal region of Kenya.

03

Methodology

Study design, data sources, variables, and spatial statistical methods



Study Design & Data Sources

Study Design

Type: Cross-sectional spatial study

Area: Coastal region of Kenya, sub-county level

Population: Children aged 9–15 years

Data: Secondary survey data aggregated to sub-county level

Geography: Sub-county boundary shapefiles from GADM database

Variables

Dependent: Bilharzia prevalence (proportion positive at sub-county level)

Independent (aggregated at sub-county):

- Mean age of children
- Proportion male/female
- Proportion using unprotected water
- Proportion without toilet access

Data Processing

- Initial dataset: **31,275** children's entries
- Filtered ages 9–15, removed missing values
- Added longitude/latitude from Kenya Subnational Boundaries
- Final sample: **26,188** children
- Variables: age, gender, county, sub-county, toilet, treated water

PREVALENCE CALCULATION

$$P_i = \frac{c_i}{N_i} \times 100 \text{ where } P_i = \text{prevalence in sub-county } i,\ c_i = \text{infected children},\ N_i = \text{total children examined}$$

Spatial Analysis Framework

1. Spatial Descriptive Analysis

- Calculate prevalence per sub-county
- Compute mean and standard deviation
- Generate **choropleth maps** to visualize spatial patterns
- Identify high and low prevalence clusters

2. Spatial Autocorrelation

- **Moran's I statistic** to test for clustering
- Spatial weights matrix (5-nearest neighbors)
- Test: clustered, dispersed, or random?
- Positive I = clustering, near-zero = random

3. Spatial Association

- **Spatial regression** analysis at sub-county level
- Model prevalence as function of demographic and environmental variables
- Multivariate logistic and linear regression

MORAN'S I FORMULA

$$I = \frac{n}{W} \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}(p_i - \bar{P})(p_j - \bar{P})}{\sum_{i=1}^{n}(p_i - \bar{P})^2}$$ where w_{ij} = 1 if sub-counties share a boundary, 0 otherwise

Software: R statistical software with spatial analysis packages

04

Results & Analysis

Spatial variation, demographic and environmental associations, multivariate models

Spatial Variation in Bilharzia Prevalence

Overall Prevalence: 6.34% among 26,188 children aged 9–15 years | Chi-squared test: x2 = 1900.2, p < 0.001 | **150-fold difference** between highest and lowest sub-counties

County	Sub-county	Total Children	Infected	Prevalence (%)	Risk Level
Tana River	Galole	797	234	29.4	Very High
Kilifi	Magarini	1,613	308	19.1	Very High
Tana River	Garsen	1,034	172	16.6	High
Kilifi	Rabai	875	87	9.94	High
Kwale	Msambweni	775	71	9.16	High
...	...	...	...	...	...
Mombasa	Changamwe	1,156	16	1.38	Low
Mombasa	Jomvu	720	9	1.25	Low
Lamu	Lamu East	140	1	0.71	Very Low
Kilifi	Kilifi North	1,446	8	0.55	Very Low
Taita Taveta	Wundanyi	763	3	0.39	Very Low
Taita Taveta	Mwatate	1,025	2	0.20	Very Low

Key Finding: The null hypothesis (H01) is rejected at p < 0.001. Prevalence shows highly significant spatial variation with clear hotspots in Tana River and northern Kilifi sub-counties.

Demographic Risk Factors

Gender Association

22.818

Chi-squared value, p = 1.78 x 10^{-6}

1.291

Adjusted OR (boys vs girls)

- Highly significant association between gender and infection
- **Boys have 29.1% higher odds** of infection (independent effect)
- Likely due to greater water contact (swimming, fishing)

Age Association

p = 0.548

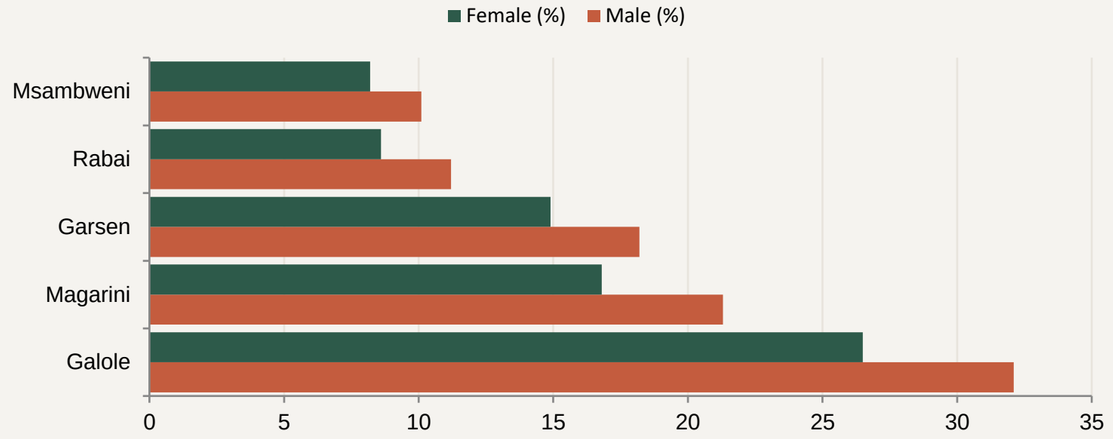
Welch's t-test (not significant)

r = -0.423

Mean age vs prevalence (p = 0.031)

- No significant individual-level age effect
- Ecological association: sub-counties with younger average age tend to have higher prevalence

Prevalence by Gender (Top Sub-counties)



H02 is rejected. Gender is a strong individual-level risk factor. Boys at higher risk due to likely greater contact with water bodies. Gender-sensitive messaging needed in health education.

Aggregate-level correlations with prevalence:

- Proportion male: r = 0.308, p = 0.126 (marginally significant)
- Mean age: r = -0.423, p = 0.031 (statistically significant)

Environmental Risk Factors

Water Source Association

213.82

Chi-squared, $p < 2.2 \times 10^{-16}$

OR 0.575

Safe water reduces odds by 42.5%

- Children using unsafe water sources had dramatically higher infection rates
- Aggregate: water source vs prevalence r = -0.290 (not significant after adjustments)

Toilet Availability — Strongest Protector

341.47

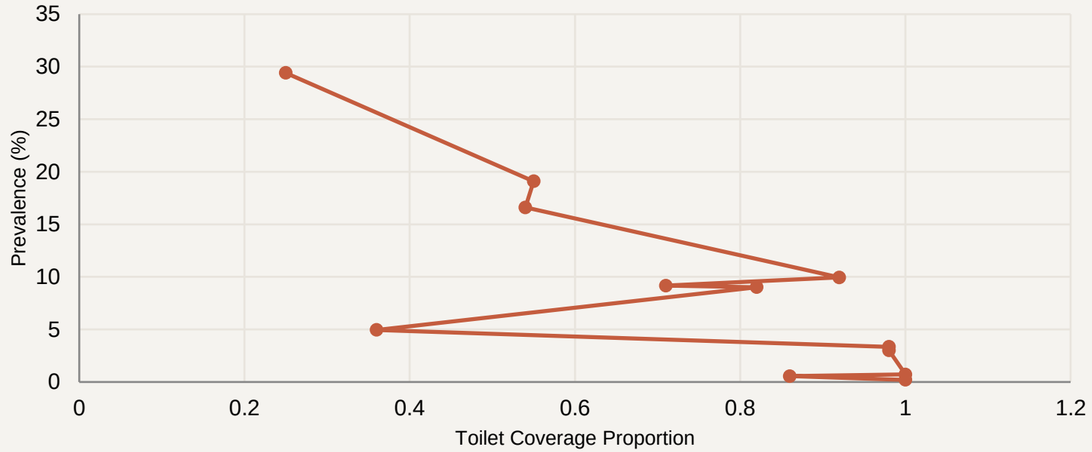
Chi-squared, $p < 2.2 \times 10^{-16}$

OR 0.442

Toilet reduces odds by 55.8%

- Absence of toilet strongly associated with infection
- Aggregate: toilet coverage vs prevalence **r = -0.614, p = 0.0008** (highly significant)

Toilet Coverage vs. Prevalence by Sub-county



H03 is rejected. Both water source and toilet availability are strongly linked to infection at individual level, while toilet coverage emerges as the dominant protective factor at sub-county level.

WASH interventions (Water, Sanitation, and Hygiene) are clearly indicated as a core control strategy for Bilharzia in coastal Kenya.

Spatial Autocorrelation in Prevalence

GLOBAL MORAN'S I TEST RESULTS

Moran's I = 0.235
Positive spatial autocorrelation

Standard Deviate = 3.293
Highly significant z-score

p-value = 0.000496
Significant at p < 0.001

Interpretation

- **Moderate but highly significant** positive spatial clustering detected
- High-prevalence sub-counties are **geographically proximate**
- Notable clusters in **Tana River** (Galole, Garsen, Bura) and **northern Kilifi** (Magarini, Rabai)
- Low-prevalence areas also cluster together (Mombasa, Taita Taveta)
- Validates the use of **spatial statistics** for targeted resource allocation

Spatial Clustering Pattern

HIGH-PREVALENCE CLUSTERS

- Galole (29.4%)
- Magarini (19.1%)
- Garsen (16.6%)
- Rabai (9.94%)
- Msambweni (9.16%)
- Matuga (9.02%)

LOW-PREVALENCE CLUSTERS

- Mwatate (0.20%)
- Wundanyi (0.39%)
- Mvita (0.41%)
- Kilifi North (0.55%)
- Lamu East (0.71%)
- Jomvu (1.25%)

H04 is rejected. Bilharzia prevalence exhibits significant positive spatial autocorrelation (Moran's I = 0.235, p = 0.0005). Interventions in one high-burden sub-county are likely to produce spillover benefits to neighbors.

Multivariate Risk Models

Multivariate Logistic Regression (Individual Level)

Variable	Adjusted OR	95% CI	p-value	Interpretation
Gender (male)	1.291	1.167–1.429	7.38e-7	29.1% higher odds
Age (per year)	1.015	0.985–1.046	0.326	No effect
Safe water source	0.575	0.518–0.638	<2.2e-16	42.5% reduction
Toilet available	0.442	0.396–0.494	<2.2e-16	55.8% reduction

Multivariate Linear Regression (Sub-county Level)

Predictor	Beta	Std. Error	p-value	Interpretation
prop_male	0.765	0.422	0.085	Marginally positive
Mean age	-0.131	0.068	0.067	Marginally negative
prop_water_1	-0.074	0.055	0.190	Not significant
prop_toilet_1	-0.170	0.062	0.012	Strongest factor

MODEL FIT

$R^2 = 0.474$

Model explains 47.4% of spatial variation in prevalence. Overall F = 6.628, p = 0.0013.

Key Interpretations

- **Toilet availability** is the single most robust protective factor at both individual (OR 0.442) and ecological levels
- **Unsafe water** has a strong direct effect but shares variance with sanitation
- **Male gender** remains an independent behavioral risk factor
- **Age shows no independent influence**

Conclusion: WASH-related factors explain more unique variance than demographics, reinforcing the spatial hotspots identified in Objective 1.

05

Conclusions & Recommendations

Key findings, implications, and evidence-based intervention strategies

Key Findings & Implications

1 Extreme Spatial Heterogeneity

Prevalence ranges from **0.195%** (Mwatate) to **29.4%** (Galole) — a 150-fold difference. Significant positive spatial autocorrelation (Moran's I = 0.235, p = 0.0005) confirms geographical clustering.

2 Gender as Independent Risk Factor

Boys have **29.1% higher odds** of infection (adjusted OR = 1.291, p < 0.001) even after controlling for age, water source, and toilet access — a behavioral risk factor.

3 Toilet Availability — Strongest Protector

Most robust protective factor at both individual (OR 0.442) and ecological levels (beta = -0.170, p = 0.012). Reduces infection odds by **55.8%** independently.

All four null hypotheses rejected. WASH factors explain more unique variance than demographics, reinforcing the need for targeted interventions.

Hypotheses Testing Summary

Hypothesis	Test Result	Decision
H01	$Chi^2 = 1900.2$, $p < 0.001$	Rejected
H02	$Chi^2 = 22.8$, $p = 1.8e-6$	Rejected
H03	$Chi^2 = 341.5$, $p < 2.2e-16$	Rejected
H04	Moran's I = 0.235, p = 0.0005	Rejected

Implications for Interventions

- Interventions in one high-burden sub-county produce **spillover benefits** to neighbors
- **Gender-sensitive** health education campaigns needed
- **WASH infrastructure** (especially sanitation) should be prioritized over MDA alone
- **Precision targeting** at sub-county level is more effective than uniform county-wide programs

Recommendations

For County and National Authorities

1. Precision targeting of hotspots

Prioritize MDA in top 8 high-prevalence sub-counties (>8%): Galole, Magarini, Garsen, Rabai, Msambweni, Matuga, Bura, Lunga Lunga

2. WASH infrastructure scale-up

Targeted sanitation campaign aiming >90% toilet coverage within 3–5 years in hotspot sub-counties

3. Surveillance and monitoring

Establish sub-county sentinel surveillance; incorporate Moran's I into routine reporting

For Communities and Residents

1. Behavior Change Communication — gender-specific messages targeting boys on safe water play

2. Community-Led Total Sanitation (CLTS) — empower residents to build latrines

3. School-Based Programs — integrate Bilharzia education into curriculum

Integrated Intervention Framework

SHORT-TERM (0–12 mo)

Intensified MDA + emergency latrine construction in hotspots

MEDIUM-TERM (1–3 yr)

Full WASH infrastructure + behavior change communication campaigns

LONG-TERM (3–5 yr)

Sustained surveillance, behavior maintenance, economic diversification

Potential impact: Reduce prevalence by >50% within 5 years in worst-affected sub-counties through MDA + WASH + behavior change.

Study Limitations

• Cross-sectional design limits causal inference • Limited to water source and toilet type • Used point centroids; ward-level shapefiles would enable finer mapping

Conclusion & Policy Recommendations

CONCLUSION

This study demonstrates that Bilharzia in coastal Kenya is a **spatially clustered, environmentally driven disease** with clear modifiable risk factors — most notably lack of toilet access. By rejecting all four null hypotheses and quantifying independent effects through multivariate modelling, the findings provide actionable intelligence for authorities and communities alike.

Key policy recommendation: Shift from uniform MDA to a **precision public health strategy** that combines targeted chemotherapy with aggressive sanitation improvement in the identified hotspots.

Policy Integration

- Embed findings into **County Integrated Development Plans (CIDPs)**
- Integrate into **National School Health Program**
- Advocate for ring-fenced funding under **Kenya NTD Master Plan**
- Use sub-county prevalence map to guide **annual MDA rounds**
- Collaboration between **national government, county authorities, and communities** is essential

Future Research Directions

- **Bayesian CAR models** for spatially explicit multivariate analysis
- **Cost-effectiveness analysis** of toilet-focused vs. MDA-only interventions
- **Qualitative studies** on barriers to toilet adoption
- Incorporate **snail density, water pH, temperature** as environmental variables
- Use **ward-level shapefiles** for finer-scale choropleth mapping

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